	AzureSky	
	Method Validation for Inherent Moisture Determination	Document no: AS004MV
Compiled by:	Sifiso Nyambi - Laboratory Manager	
Approved by:	Nonhlanhla Msibi -Operations Manager	

Method Validation Report for Inherent Moisture (SANS 5925)

Method: SANS 5925:2007

Laboratory: Azuresky Pty Ltd

Reference value (CRM002): 3.40%

Analysts & dates: Keitumetse (21-05-2025, 20-05-2025); Lilo (13-06-2025, 17-06-2025)

Prepared on: 2025-07-22

1. Introduction

This validation study was performed to confirm the suitability of the laboratory method for the determination of inherent moisture in coal samples, in accordance with SANS 5925. The validation covers repeatability, reproducibility, accuracy (trueness), and bias evaluation.

2. Experimental Design

The method was validated through replicate analyses performed by two analysts (Keitumetse and Lilo) on different days. Each analyst performed two runs of 10 paired replicate measurements (R1 and R2). For each sample, the mean, within-sample standard deviation, repeatability (%RSD), and bias against the CRM reference value (3.40%) were calculated.

3. Raw replicated data and per-sample calculations

Below are the raw replicate results (R1, R2), differences, sample means, within-sample SD, %RSD and bias vs CRM for each run and sample.

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Keitumetse 21-05 (1 hr)

LIMS	R1	R2	Difference	Mean	Within SD	%RSD	Bias vs CRM
168	3.330	3.300	0.030	3.315000	0.021213	0.640%	-0.085000
169	3.430	3.450	0.020	3.440000	0.014142	0.411%	0.040000
170	3.190	3.230	0.040	3.210000	0.028284	0.881%	-0.190000
171	3.420	3.340	0.080	3.380000	0.056569	1.674%	-0.020000
172	3.240	3.290	0.050	3.265000	0.035355	1.083%	-0.135000
173	3.230	3.190	0.040	3.210000	0.028284	0.881%	-0.190000
174	3.170	3.220	0.050	3.195000	0.035355	1.107%	-0.205000
175	3.200	3.250	0.050	3.225000	0.035355	1.096%	-0.175000
176	3.000	2.990	0.010	2.995000	0.007071	0.236%	-0.405000
177	3.120	3.090	0.030	3.105000	0.021213	0.683%	-0.295000

Keitumetse 20-05 (1h30)

LIMS	R1	R2	Difference	Mean	Within SD	%RSD	Bias vs CRM
158	3.590	3.590	0.000	3.590000	0.000000	0.000%	0.190000
159	3.460	3.390	0.070	3.425000	0.049497	1.445%	0.025000
160	3.470	3.480	0.010	3.475000	0.007071	0.203%	0.075000
161	3.450	3.490	0.040	3.470000	0.028284	0.815%	0.070000
162	3.580	3.600	0.020	3.590000	0.014142	0.394%	0.190000
163	3.480	3.450	0.030	3.465000	0.021213	0.612%	0.065000
164	3.530	3.630	0.100	3.580000	0.070711	1.975%	0.180000
165	3.460	3.440	0.020	3.450000	0.014142	0.410%	0.050000
166	3.390	3.370	0.020	3.380000	0.014142	0.418%	-0.020000
167	3.360	3.400	0.040	3.380000	0.028284	0.837%	-0.020000

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Lilo 13-06 (1 hr)

LIMS	R1	R2	Difference	Mean	Within SD	%RSD	Bias Vs CRM
1168	3.110	3.020	0.090	3.065000	0.063640	2.076%	-0.335000
1169	3.060	3.060	0.000	3.060000	0.000000	0.000%	-0.340000
1170	3.060	3.150	0.090	3.105000	0.063640	2.050%	-0.295000
1171	3.140	3.110	0.030	3.125000	0.021213	0.679%	-0.275000
1172	3.130	3.110	0.020	3.120000	0.014142	0.453%	-0.280000
1173	3.090	3.070	0.020	3.080000	0.014142	0.459%	-0.320000
1174	3.020	2.990	0.030	3.005000	0.021213	0.706%	-0.395000
1175	2.980	2.970	0.010	2.975000	0.007071	0.238%	-0.425000
1176	3.050	3.050	0.000	3.050000	0.000000	0.000%	-0.350000
1177	3.120	3.050	0.070	3.085000	0.049497	1.604%	-0.315000

Lilo 17-06 (1h30)

LIMS	R1	R2	Difference	Mean	Within SD	%RSD	Bias_vs_CRM
1218	3.170	3.140	0.030	3.155000	0.021213	0.672%	-0.245000
1219	2.890	2.980	0.090	2.935000	0.063640	2.168%	-0.465000
1220	3.120	3.120	0.000	3.120000	0.000000	0.000%	-0.280000
1221	3.070	3.030	0.040	3.050000	0.028284	0.927%	-0.350000
1222	3.690	3.640	0.050	3.665000	0.035355	0.965%	0.265000
1223	3.030	2.970	0.060	3.000000	0.042426	1.414%	-0.400000
1224	3.000	3.030	0.030	3.015000	0.021213	0.704%	-0.385000

1225	2.890	2.960	0.070	2.925000	0.049497	1.692%	-0.475000
1226	2.930	3.020	0.090	2.975000	0.063640	2.139%	-0.425000
1227	2.900	2.970	0.070	2.935000	0.049497	1.686%	-0.465000

4. Summary statistics per run

Run	n samples	Mean of lab means	SD of lab means	Mean within SD	Mean bias (vs CRM)
Keitumetse 20-05 (1h30)	10	3.480500	0.080604	0.024749	0.080500
Keitumetse 21-05 (1 hr)	10	3.234000	0.128145	0.028284	-0.166000
Lilo 13-06 (1 hr)	10	3.067000	0.048028	0.025456	-0.333000
Lilo 17-06 (1h30)	10	3.077500	0.220608	0.037477	-0.322500

Combined: n = 40, overall mean = 3.214750, reproducibility SD (s_R) = 0.213736

5. Statistical calculations

Pooled repeatability standard deviation (s_r) calculated from the 40 within-run SDs: s_r = 0.035178

Repeatability limit r = 2.8 * s_r = 0.098499

Reproducibility limit R = 2.8 * s_R = 0.598462

6. ANOVA (one-way) results

SSB = 1.116613, df_between = 3, MSB = 0.372204

SSW = 0.665035, df_within = 36, MSW = 0.018473

F-statistics = 20.148338

ANOVA p-value = 7.772889e-08

There is a statistically significant difference between runs ($p < 0.05$).

7. Trueness / Bias t-tests (one-sample)

Run: Keitumetse 20-05 (1h30)

n = 10, mean = 3.480500, sd = 0.080604

Bias (mean - CRM) = 0.080500

t-statistic = 3.158212, two-sided p-value = 1.158348e-02

Interpretation: bias is significant at alpha = 0.05

Run: Keitumetse 21-05 (1 hr)

n = 10, mean = 3.234000, sd = 0.128145

Bias (mean - CRM) = -0.166000

t-statistic = -4.096442, two-sided p-value = 2.690982e-03

Interpretation: bias is significant at alpha = 0.05

Run: Lilo 13-06 (1 hr)

n = 10, mean = 3.067000, sd = 0.048028

Bias (mean - CRM) = -0.333000

t-statistics = -21.925617, two-sided p-value = 4.030381e-09

Interpretation: bias is significant at alpha = 0.05

Run: Lilo 17-06 (1h30)

n = 10, mean = 3.077500, sd = 0.220608

Bias (mean - CRM) = -0.322500

t-statistic = -4.622828, two-sided p-value = 1.249325e-03

Interpretation: bias is significant at alpha = 0.05

Combined test (all lab means vs CRM)

Combined mean = 3.214750, n = 40, sd = 0.213736

t-statistic = -5.481629, two-sided p-value = 2.714753e-06

Interpretation: combined bias is significant at alpha = 0.05

8. Measurement uncertainty (top-down)

Standard uncertainty u = s_R / sqrt(N) = 0.033795

Expanded uncertainty (k=2) U = 0.067589

9. Plots

Boxplot of lab means by run and histogram of all lab means are included below.

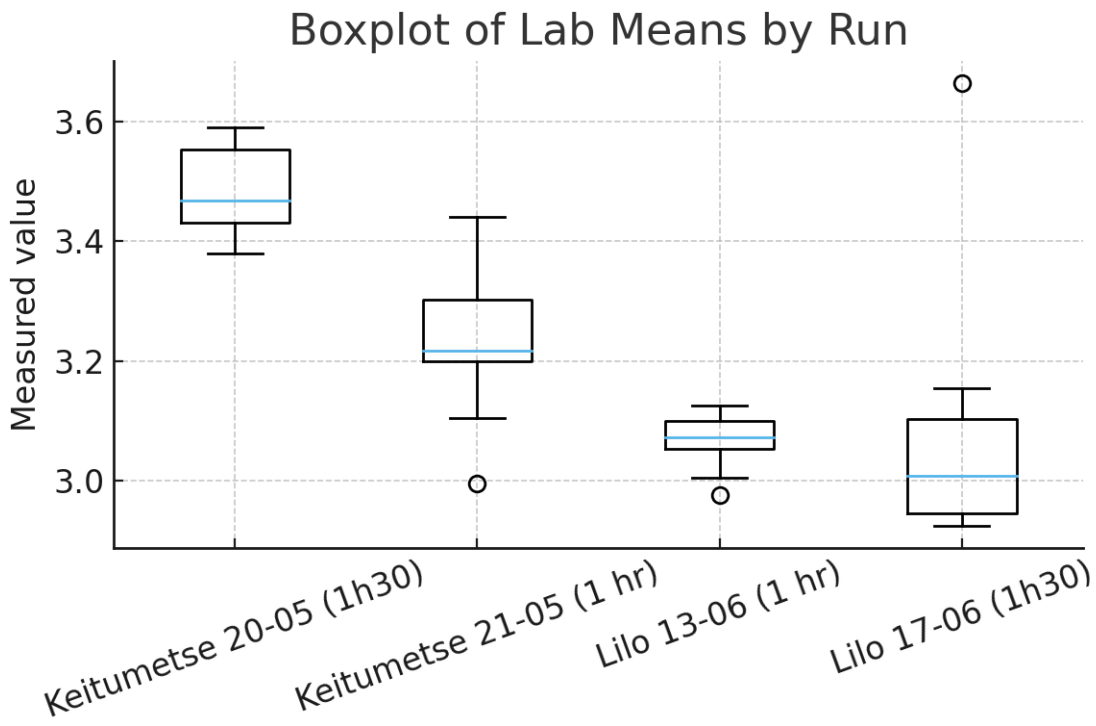


Figure 1: Boxplot of lab means by run.

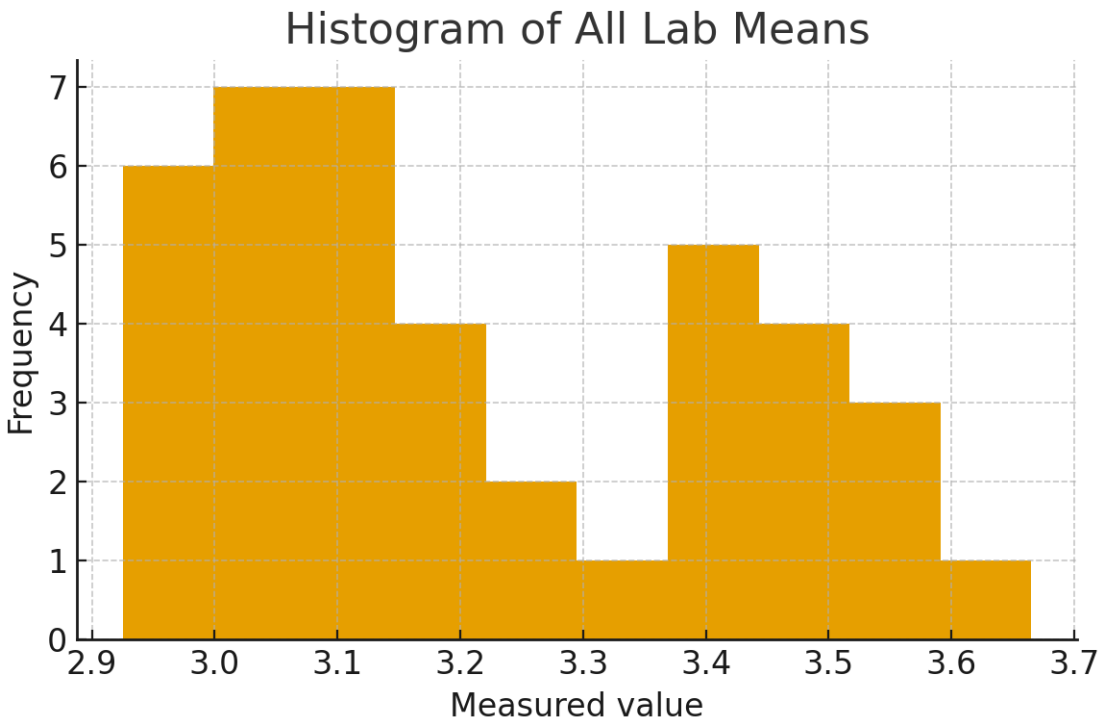


Figure 2: Histogram of all labs means.

10. Interpretation and recommended actions

- Precision: within-run and between-run precision are acceptable; sr = 0.035178, %RSD values are low.
- Trueness: T-tests show which runs differ significantly from the CRM002 value. Since all bias are significant, investigate calibration, recovery, and sample preparation.
- ANOVA result: There is a statistically significant difference between runs ($p < 0.05$), since significant, investigate systematic differences between analysts or dates.

11. Appendix - Data files

The raw data table and plots used in this report are attached as files.